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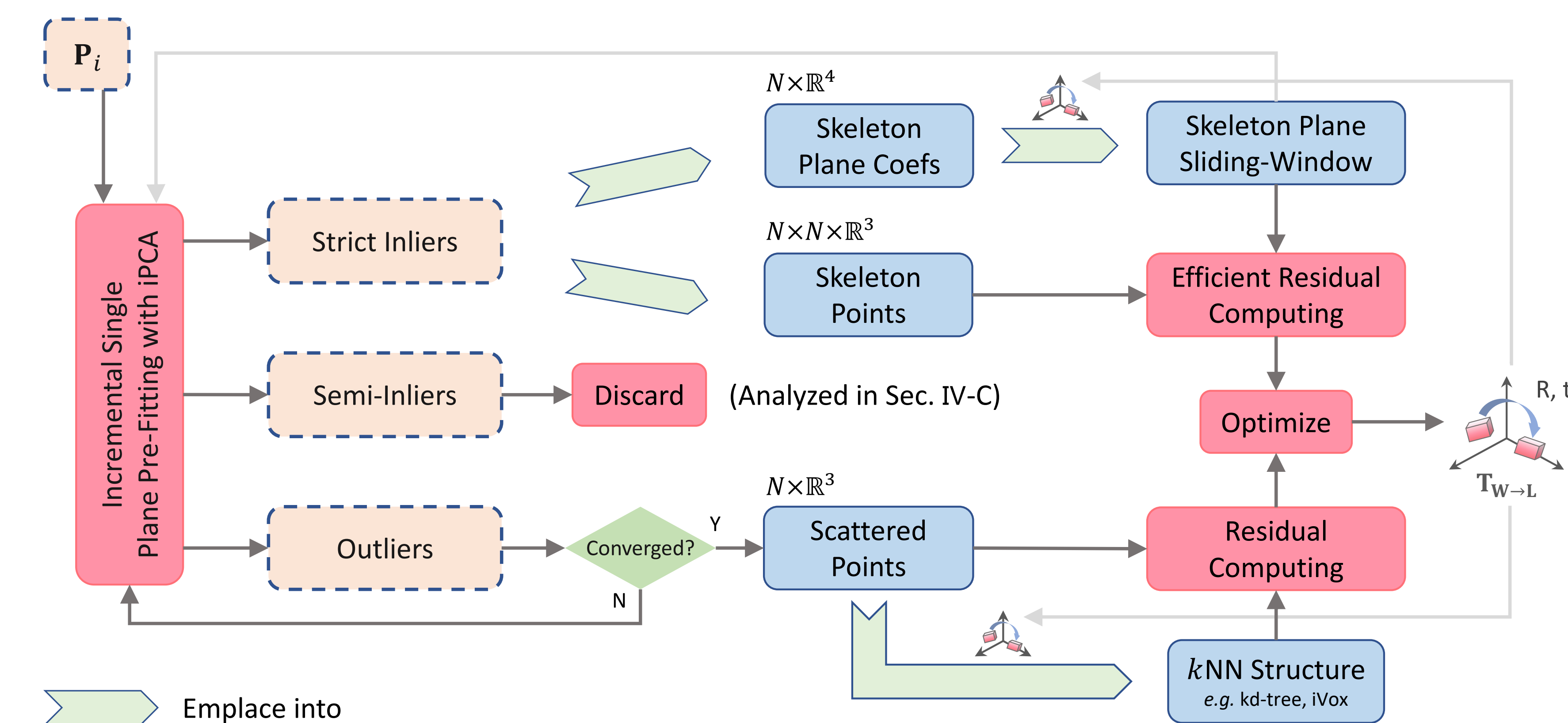
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LIO-PPF: Fast LiDAR-Inertial Odometry via Incremental Plane Pre-Fitting and Skeleton Tracking

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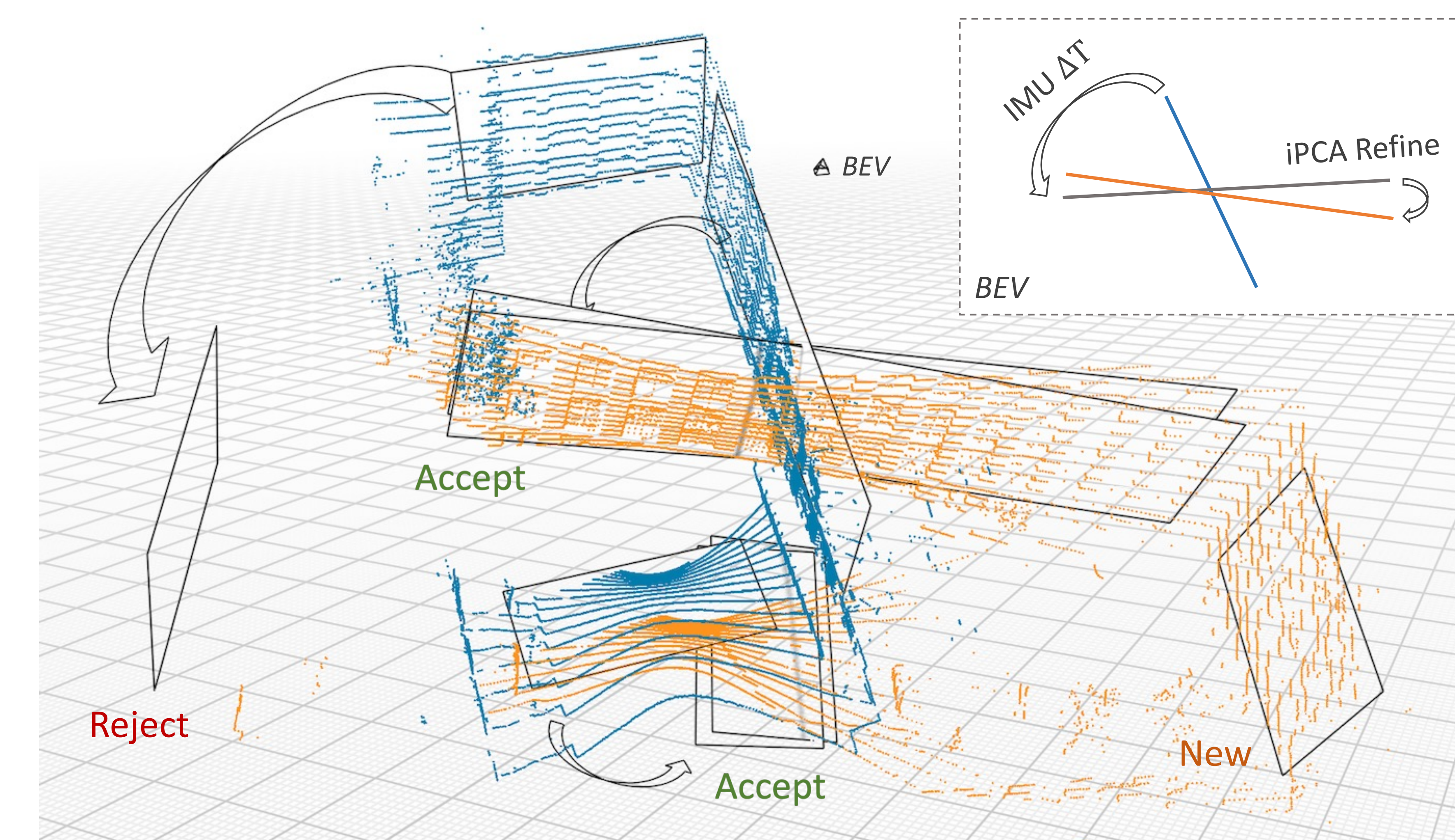
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Pipeline

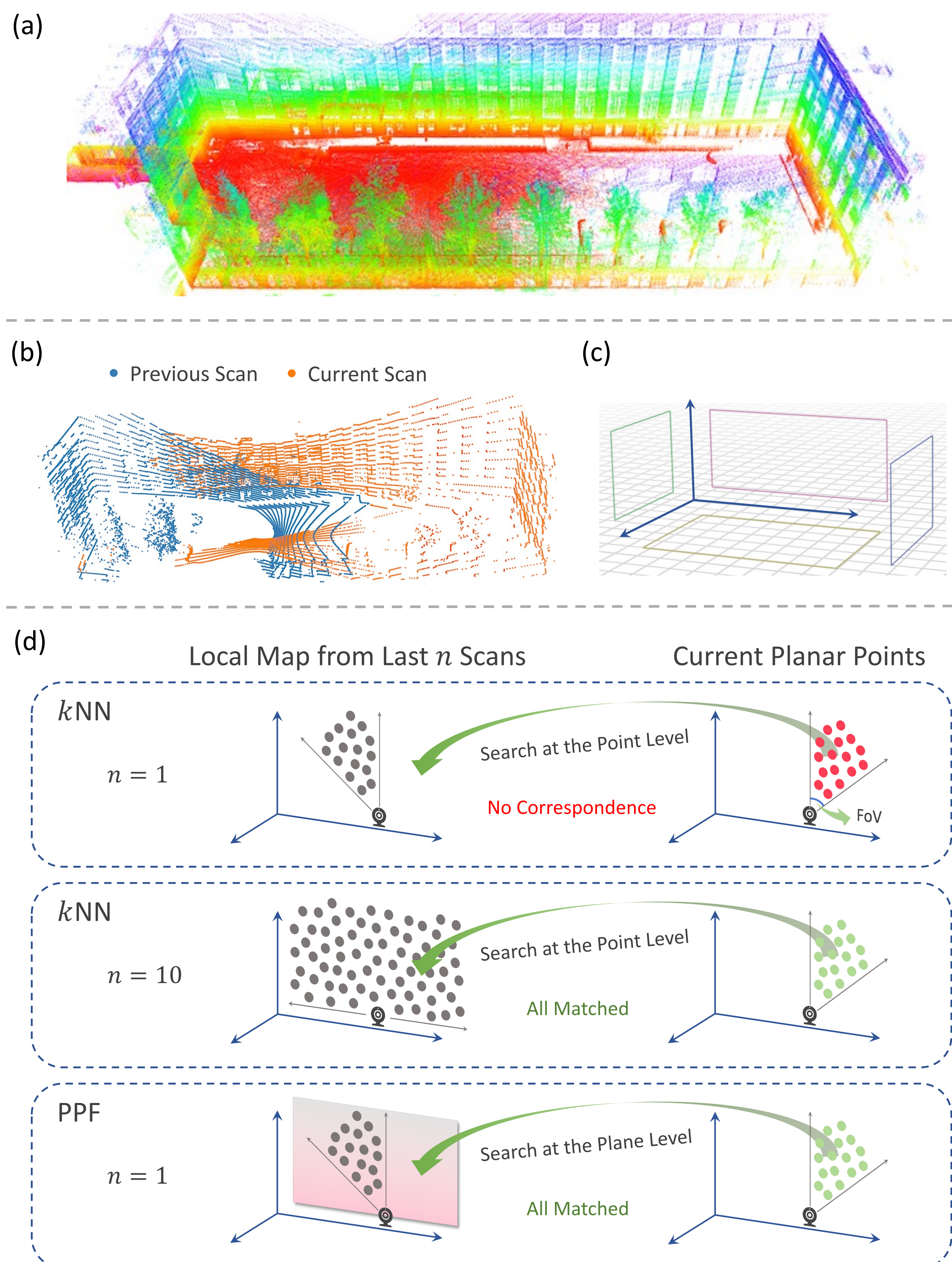


Incremental Plane Fitting

- Planes are not fitted individually for each scan, let alone for each point, but are updated **incrementally** as the scene **'flows'**.
- Planes of last scan (after the IMU ΔT transformation) are treated as **initial plane guess**.
- The **accepted** guesses need only to be refined.
- Only a few planes are **newly** fitted from scratch.

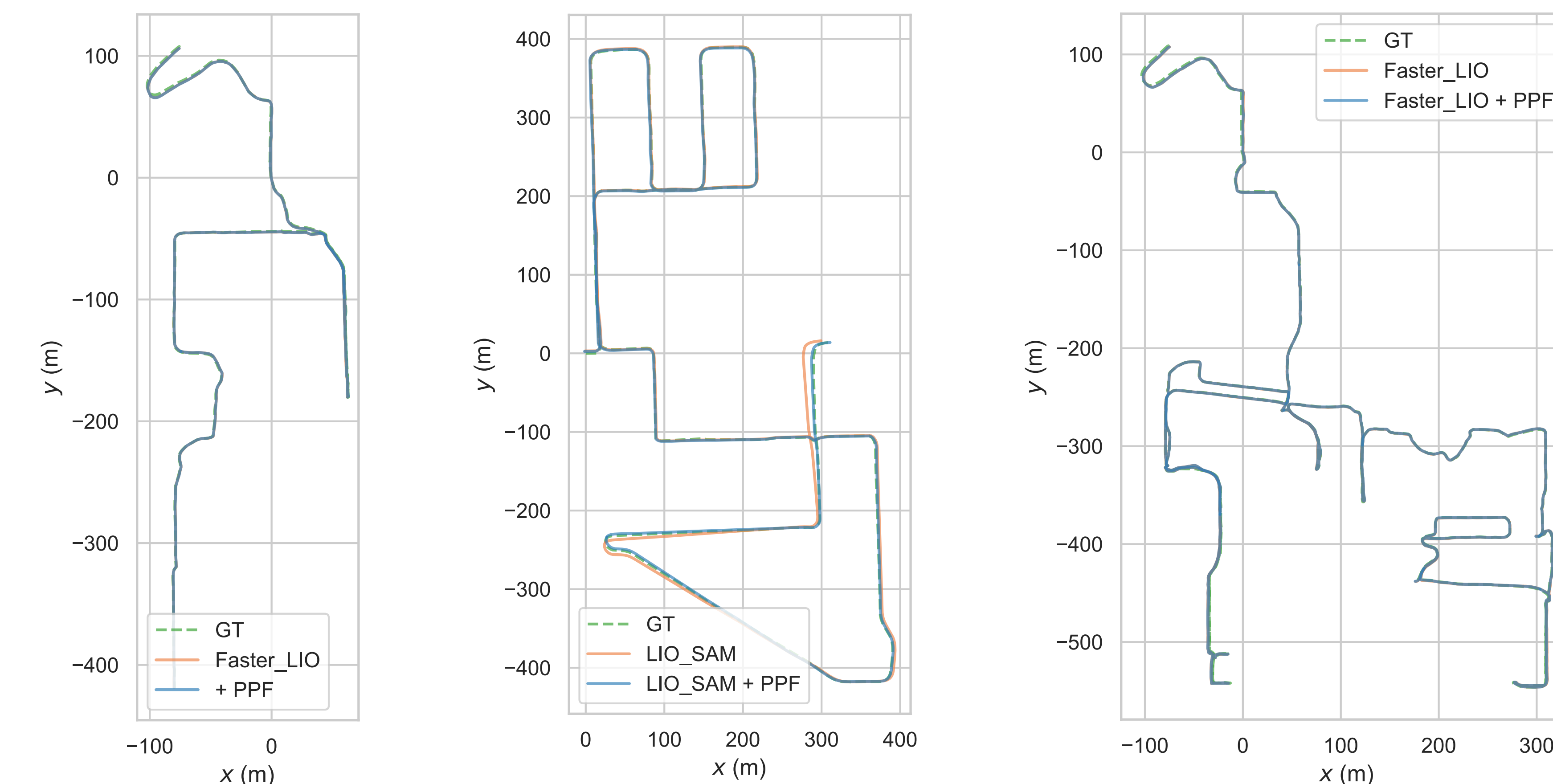


Motivation



- Large planes across the scans form the **basic skeleton** of the scene and reveal its overall geometry structure.
- A large local map is needed by kNN otherwise most points can't find neighbors to fit planes.
- While we use the **basic skeleton** (Fig. c) to represent the scene for point matching. The search field is enlarged to plane level with no kNN search required any longer.

Estimated Trajectories



Time & Memory Cost

TIME (ms) AND MEMORY CONSUMING OF EACH STEP IN LIO-SAM [5] USING kNN OR PPF

Dataset	PPF		Build Local Map		Build kd-tree		Scan Match		Calc. Residual		Total time ¹		Local Map Size ²	
	kNN	PPF	kNN	PPF	kNN	PPF	kNN	PPF	kNN	PPF	kNN	PPF	kNN	PPF
LIOSAM-3	0	2.64	21.28	10.48	8.43	5.51	19.97	13.84	1.50	0.69	45.12	30.80	66.9k	44.1k
LIOSAM-4	0	3.16	28.12	13.65	9.56	6.25	21.99	15.96	1.66	0.76	54.36	36.98	75.8k	49.9k
LIOSAM-5	0	3.07	18.20	9.97	8.55	6.05	22.03	17.51	2.01	0.97	44.37	34.69	67.9k	48.3k
LIOSAM-2	0	3.61	17.69	10.33	8.88	6.36	23.36	18.55	2.11	1.14	44.95	36.39	70.4k	50.5k
KITTI-7	0	1.30 [†]	55.08	21.56	8.20	3.72	42.93	19.47	3.95	0.83	112.66	58.73	67.6k	29.4k
KITTI-2	0	1.96 [†]	36.76	13.95	5.19	2.61	35.63	16.99	3.80	0.99	91.25	51.77	41.6k	15.7k
KITTI-3	0	1.50 [†]	44.55	24.30	4.30	2.75	28.96	17.48	2.49	0.78	93.05	62.82	37.0k	19.5k
KITTI-4	0	1.25 [†]	45.86	21.92	4.37	2.32	28.47	15.23	2.95	0.73	93.96	58.02	37.3k	17.0k
KITTI-6	0	1.48 [†]	43.29	17.32	5.48	2.34	30.96	15.15	4.12	0.90	93.23	52.93	44.3k	15.9k
KITTI-5	0	1.22 [†]	55.24	23.11	7.31	3.44	43.17	21.10	3.52	0.73	114.89	64.43	60.8k	26.6k
KITTI-1	0	1.29 [†]	45.04	22.79	4.70	2.69	32.25	17.98	3.07	0.80	97.18	61.96	40.3k	19.8k